



OUTPROMPT

THE AI STRATEGY CHALLENGE

PROBLEM 4:

SMALLHOLDER AGRI - OWNING THE MARKET LINKAGE

ROUND 1 SUBMISSION BRIEF

The Situation

You advise a **Farmer Producer Organization (FPO)** - a member-owned collective of smallholder farmers - that wants to lift its members' incomes by **owning the link between farm and market**. Today, smallholders get poor prices and weak advice: they sell at the farmgate to a chain of intermediaries, capture only a fraction of what the consumer finally pays, and farm largely blind to what the market wants. The FPO can build an **advisory-plus-linkage** model - telling farmers what and when to grow, then aggregating and selling their produce closer to the end buyer - capturing some of the value the middle of the chain takes today.

But it must decide what kind of thing it is building. Is this a **profitable business** that can stand on its own unit economics, or an **impact play** that needs subsidy or grant to survive? And where should it start - which crops, which geographies, which farmers?

Why This Is a Massive Opportunity

India has roughly **146 million farm holdings**, **~86% of them small and marginal (under 2 hectares)**, with an average farm size of just **~1.08 hectares** - and shrinking. These millions of tiny, scattered farms are exactly why the middle of the chain is so powerful: no single farmer has the volume, storage, information, or reach to sell beyond the local intermediary. Yet smallholders are not marginal to output - on just **~44% of the land** they produce **~70% of India's vegetables**, **~55% of fruits**, and **~52% of cereals**.

For many crops the farmer captures only a minority of the final consumer price; the rest accrues to aggregators, transporters, commission agents, and retailers. The World Economic Forum has estimated that technology across the farmgate-to-fork chain could add **\$62–76 billion a year** to Indian agriculture, with 50–60% of that potentially captured by farmers. The question is whether a single FPO can actually capture a slice of this - and make money doing so.

Globally, smallholders matter enormously: the FAO estimates they supply roughly **70% of all food produced**, yet many remain food-insecure themselves, and more than **80% of smallholders sell into local and domestic markets** rather than export - so the prize for this FPO is overwhelmingly the home market, not foreign buyers.

The Tension at the Heart of This Case

Cutting out intermediaries sounds like pure gain, but those intermediaries do real work: aggregation, transport, credit, price risk, sorting, last-mile reach across a country with too few markets. The FAO is blunt that smallholders face high transaction costs, poor infrastructure, weak market-information systems, and food loss from inadequate post-harvest handling - the very frictions the middlemen absorb. An FPO that takes over the linkage takes over those costs and risks too - on tiny volumes, with thin balance sheets and limited management capacity.

India already has 1,000-plus producer companies, and the evidence is that those staying at “crop planning plus input supply plus primary marketing” underperform - the ones that realised higher returns moved **further up the chain** into direct retailing, value addition, storage, and processing. There may be value in moving up, but so is the capability, capital, and risk an FPO may not have. The crops where farmers lose the most to intermediaries (perishables) are also the hardest and riskiest to handle.

The Decision

Which part of the value chain should the FPO own to create the greatest farmer income uplift while remaining commercially viable?

The chain runs from input supply → crop advisory → aggregation → grading & storage → transport & logistics → processing / value addition → direct sale to buyer, retailer, or consumer. The FPO cannot own all of it. Recommend **which link (or links) it should own first** - the one that lifts farmer income most while the FPO can actually resource it and make the economics work.

Answering it well means working through what it takes to defend that choice:

- **Where, concretely:** the crop and context that make your chosen link the right one (a perishable and a grain give different answers).
- **What owning it takes:** the capability, capital, and risk the FPO must absorb to own that link - and whether it realistically can.
- **Whether the math closes, for both sides:** a directional view that farmer income rises and the FPO's own unit economics are viable - not impact at the client's expense, nor margin at the farmer's.

One angle worth weighing: women make up ~43% of the agricultural labour force in developing countries, and the FAO estimates that closing their access gap to inputs and services could lift their farm yields 20–30%.

Exhibits

Exhibit A - Farmer's share of the consumer rupee, by commodity type

Commodity type	Farmer's share of consumer rupee	Implied intermediary + retail share
Foodgrains (limited processing)	~45–69%	~31–55%
Perishables (fruits & vegetables)	~30–50%	~50–70%

Exhibit B - India's farm structure: small, scattered, shrinking

India farm structure	Figure
Total operational landholdings	~146 million
Share that are small & marginal (<2 ha)	~86%
Average farm size	~1.08 ha (2.3 ha in 1970-71)
Share of farmed area operated by small & marginal	~47%

Exhibit C - Why market access is so hard (and so contested)

Market-access friction	Figure
Regulated wholesale market density (actual)	~1 per 496 sq km
Density recommended (Natl. Commission on Farmers)	~1 per 80 sq km
Share of commodity trade handled by eNAM as of 2021	<1%

Exhibit D - What the FPO could do with public data. Several live public sources exist that an advisory-plus-linkage model could build on: **Agmarknet** (daily mandi prices by crop and market), **IMD** (weather forecasts), **ISRO Bhuvan / Sentinel-2** (satellite imagery for crop and acreage monitoring), **soil-health-card data**, **MSP** (minimum support prices), and **eNAM** (the national e-market). How you'd combine them is up to you.

Exhibit E - Global context (frames the opportunity; India specifics decide the answer)

Global smallholder context (FAO)	Figure
Share of all food production from smallholders	~70%
Smallholders selling into local/domestic markets	>80%
Women as share of agri labour force (developing world)	~43%
Potential yield lift if women had equal input access	+20–30%

Exhibit F - India institutional evidence: aggregation works, and moving up the chain works better

India institutional evidence	Figure / example
Net income uplift, contract farming vs independent (Kumar 2016)	~+81%
Producer companies in India	1,000+
Farmer-consumer markets established	350+ (Rythu Bazar, Apni Mandi, Uzhavar Santhai)
FPC exemplars that moved up the chain	Sahyadri (Nashik), MAHAGRAPHES, HOPCOMS

Data You Can Explore

Feel free to use public sources as required – however, ensure you cite them properly so we can go back and verify that these numbers are correct.

What Round 1 Is - and Isn't

Working with AI is part of the job. Modern consulting increasingly expects you to work alongside AI, so that is the skill we are testing here. We are not looking for an answer solved unaided - we want to see how well you direct an AI towards a sharp recommendation, and how well you recognise where it falls short. So please do use AI to develop your response; it is expected, not optional.

Round 1 is analysis, not a build. You are not building a working prototype - that is Round 2. Concept sketches, mock-ups, or described experiences are welcome, but the deliverable is your thinking on the page.

Your Round 1 answer is **not binding**. If Round 2 changes your view, change it with no penalty - the two rounds are scored independently.